

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTESC304 - COMPUTER ARCHITECTURE & OPERATING SYSTEMS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

1 The fundamental unit of data in a computer is:

- a) Byte
- b) Bit
- c) Word
- d) Nibble

GPGPU stands for:

- a) General-Purpose Graphics Processing Unit
- b) General-Purpose Graphical Processing Unit
- c) Great Performance Graphics Processor Unit
- d) General Performance Graphic Processing Unit

3 What is the main difference between RISC and CISC architecture?

- a) Clock speed
- b) Number of registers
- c) Number of instructions
- d) Operating voltage

4 RAID stands for:

- a) Random Access Independent Disks
- b) Redundant Array of Inexpensive Disks
- c) Repeated Access Integrated Devices
- d) None of the above

5 MESI protocol is used in:

- a) Virtual memory
- b) Cache coherence
- c) Page replacement
- d) Instruction scheduling

6 Which of the following is not a function of an operating system?

- a) Process management
- b) File management
- c) Data mining
- d) Memory management

- 7 Which component performs arithmetic and logic operations?
 a) Control Unit
 b) ALU
 c) Registers
 d) Cache
- 8 Which format is used to represent floating point numbers in computers?
 a) IEEE 754
 b) ASCII
 c) BCD
 d) UTF-8
- 9 Which memory type allows simultaneous read and write operations?
 a) RAM
 b) ROM
 c) PROM
 d) EPROM
- 10 Hardwired control unit is typically:
 a) Faster but less flexible
 b) Slower and flexible
 c) Always microprogrammed
 d) Only used in GPUs
- 11 The part of OS that loads first into memory is:
 a) Shell
 b) Kernel
 c) Scheduler
 d) File System
- 12 In paging, each process is divided into:
 a) Pages
 b) Blocks
 c) Sectors
 d) Nodes

Q2. Solve the following.

- A) Describe the structure and function of a CPU with the help of a labeled diagram. 6
 B) Discuss the types of operations and operands supported by instruction sets. 6

Q3. Solve the following.

- A) Describe the organization and working of semiconductor main memory with a diagram. 6
 B) Describe the process of memory allocation using: 6
 1. Continuous memory allocation
 2. Fixed and variable partitions
 3. Fragmentation and compaction

Q4. Solve Any Two of the following.

- A) Explain the steps involved in micro-instruction sequencing and execution. 6
 B) Write a short note on multiple processor organization and its types. 6

C) Compare and contrast:

6

1. Programmed I/O
2. Interrupt-driven I/O
3. DMA

Q5. Solve Any Two of the following.

A) Explain the types of Operating Systems with examples. Also discuss the structure of an Operating System. 6

B) Explain multiple processor scheduling and real-time scheduling with examples. 6

C) Write a detailed note on System Design and Implementation of Operating Systems. 6

Q6. Solve Any Two of the following.

A) What is Process Synchronization? Explain the Critical Section Problem and outline software solutions for two or more processes. 6

B) Write detailed notes on the following: 6

1. Semaphores
2. Monitors

C) Discuss in detail the methods for handling deadlocks. 6

*** End ***